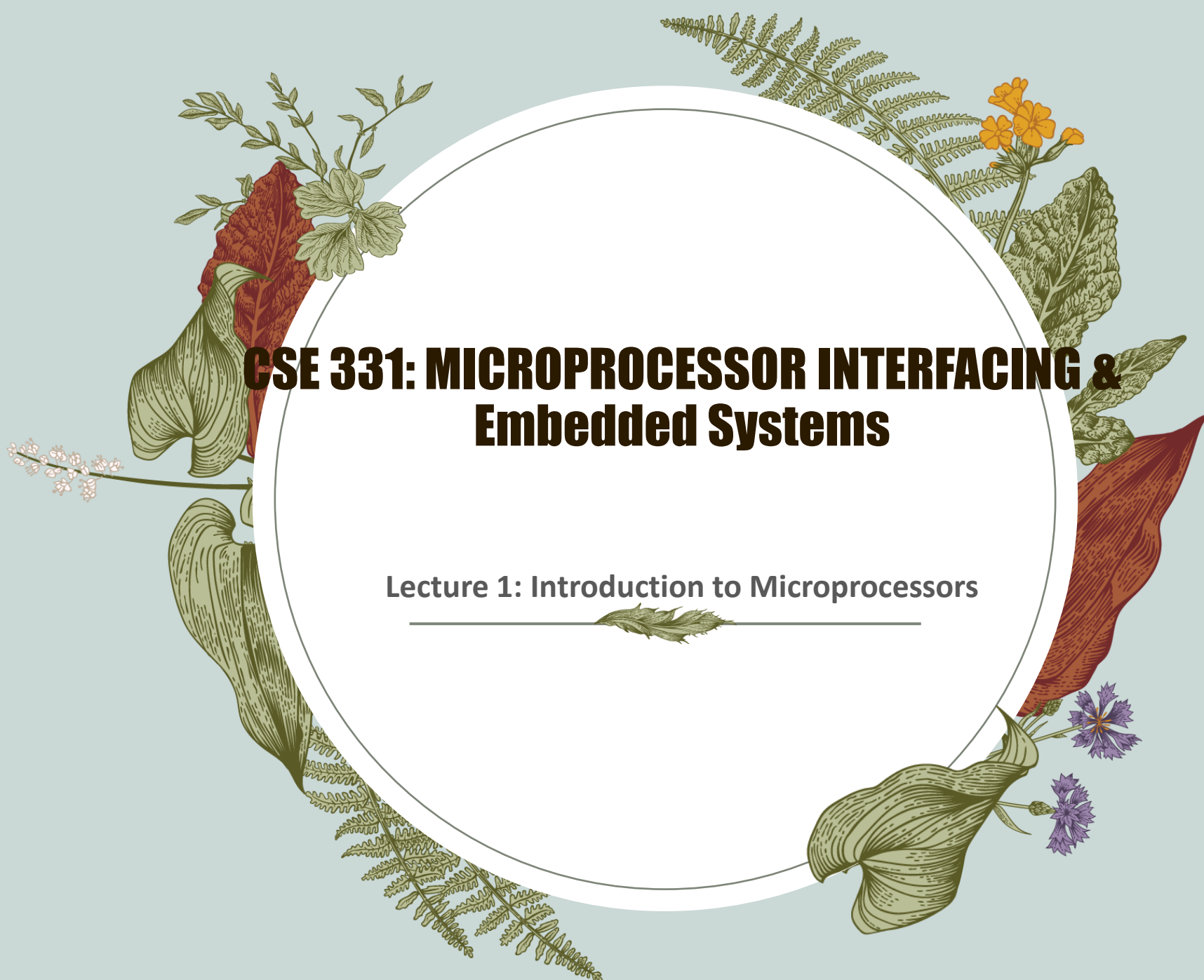


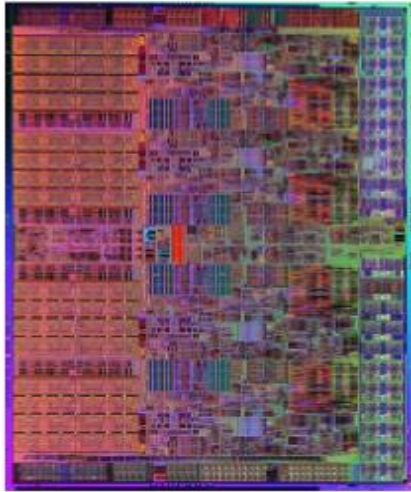
A detailed botanical illustration wreath composed of various plants, including green ferns, red maple leaves, yellow marigolds, purple cornflowers, and green bell-shaped flowers, arranged in a circular pattern around a central white circle.

CSE 331: MICROPROCESSOR INTERFACING & Embedded Systems

Lecture 1: Introduction to Microprocessors

WHAT IS A MICROPROCESSOR?

- Microprocessor is a Central Processing Unit (CPU) on a single chip.
- It contains billions of transistors connected by wires.

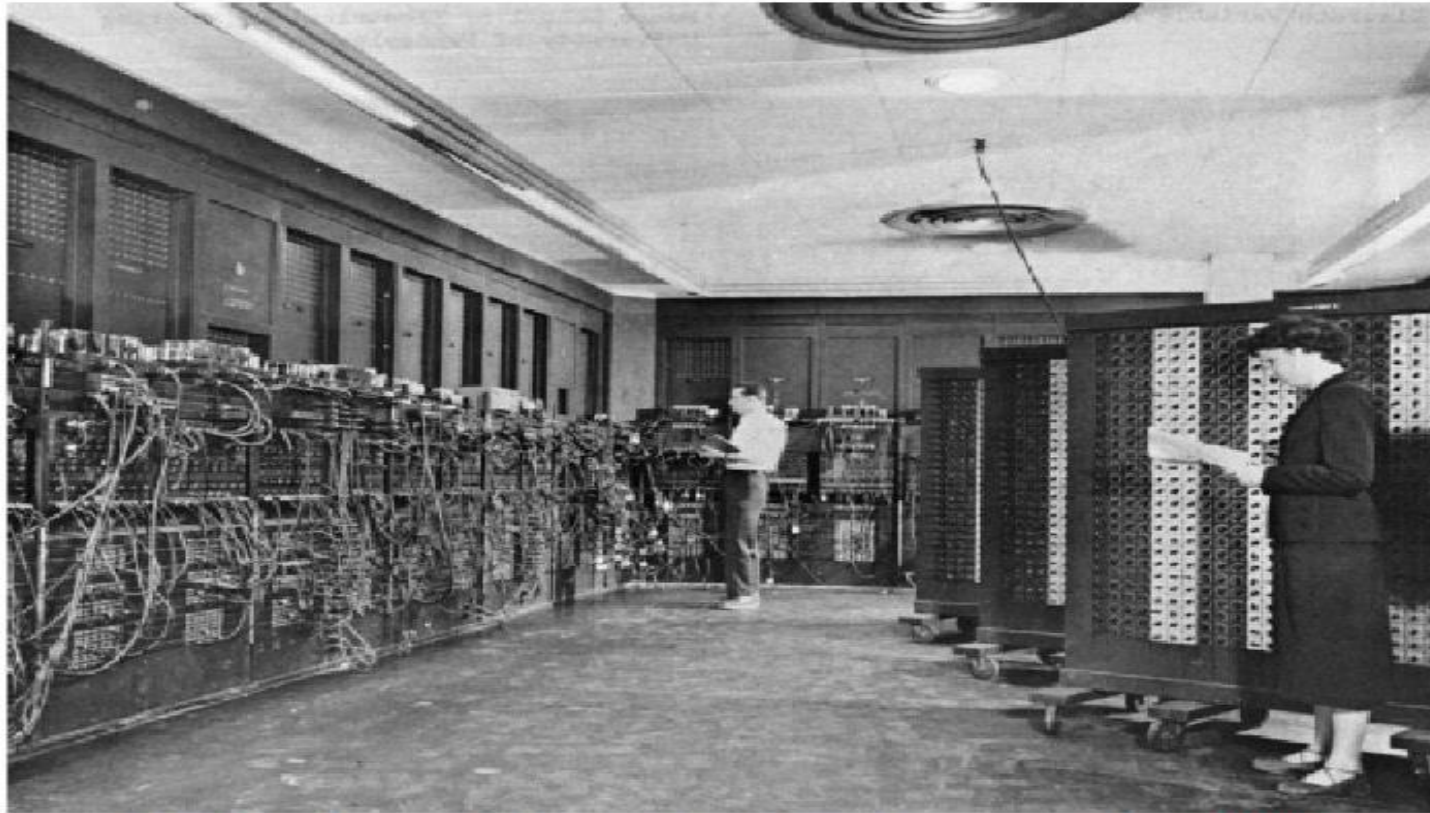


Core i7 die



Core i7 in package

ELECTRICAL NUMERICAL INTEGRATOR AND CALCULATOR



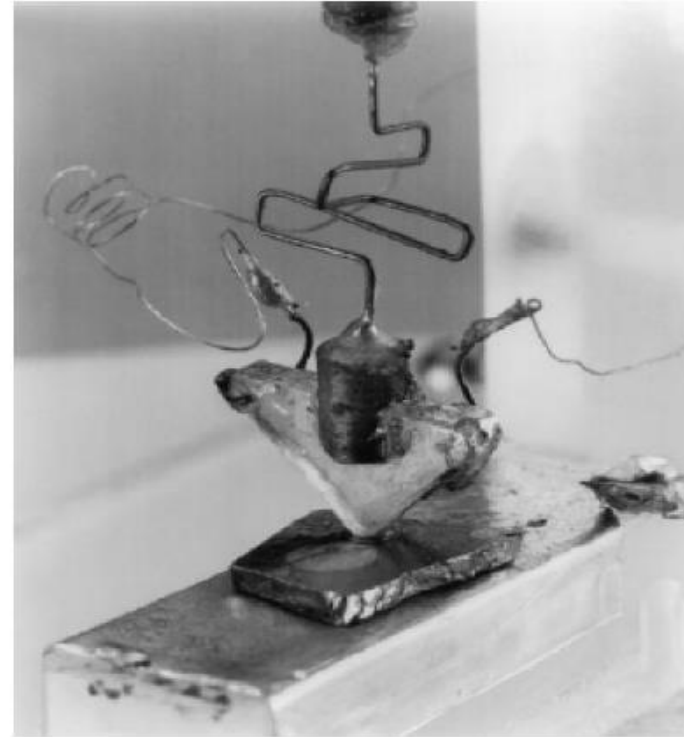
Glen Beck and Betty Snyder program the ENIAC
in BRL building 328. (Picture: U.S. Army)

ENIAC

- ☐ Designed for the U.S.Army's Ballistic Research Laboratory
- ☐ Built out of
 - ☐ 17,468 vacuum tubes
 - ☐ 7,200 crystal diodes
 - ☐ 1,500 relays
 - ☐ 70,000 resistors
 - ☐ 10,000 capacitors •
- ☐ Consumed 150 kW of power •
- ☐ Took up 72 sq m.
- ☐ Weighted 27 tons •
- ☐ Suffered a failure on average every 6 hours

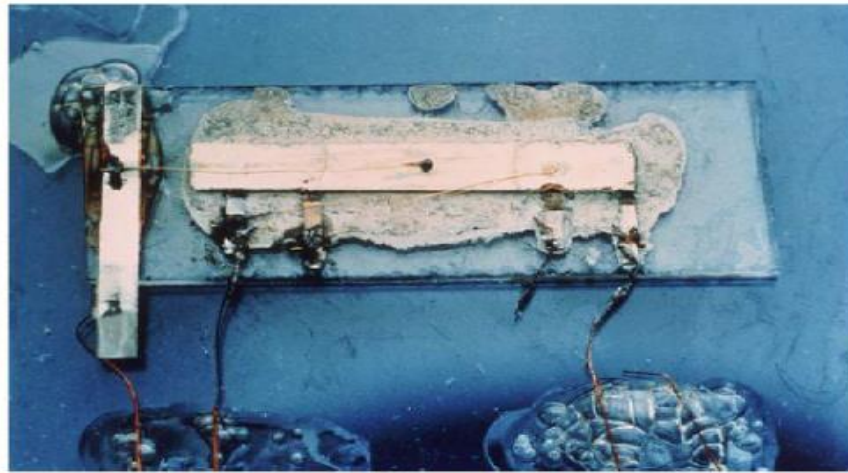
FIRST TRANSISTOR INVENTED IN 1947

- ❑ Created by a team lead by
- ❑ William Shockley at Bell Labs
- ❑ Shockley later shared the
- ❑ Noble prize in physics
- ❑ Shockley semiconductors was
- ❑ founded in Palo Alto in 1955
- ❑ In 1957 Bob Noyce, Gordon Moore, and 6 others (“Traitorous Eight”) leave to found Fairchild semiconductor



FIRST INTEGRATED CIRCUIT (IC) INVENTED IN 1959

- Proposed independently by Bob Noyce at Fairchild and Jack Kilby at Texas Instruments
- In 1968 Noyce and Moore left Fairchild to found Intel
- First IC contained a single transistor and supporting components

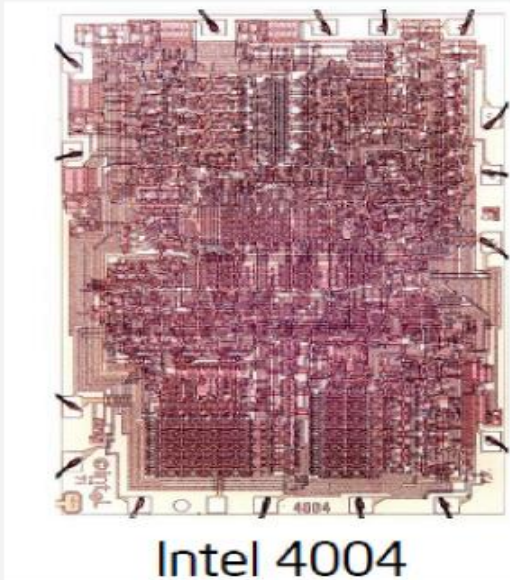


The first working integrated circuit (1.6×11.1 mm)

Image: TI

FIRST COMMERCIAL PROCESSOR BY INTEL: 4004

- Intel Introduced the 4004 in 1971
- 4004 contained 2,300 transistors
- Similar processing power as ENIAC



Intel 4004

Image: Intel

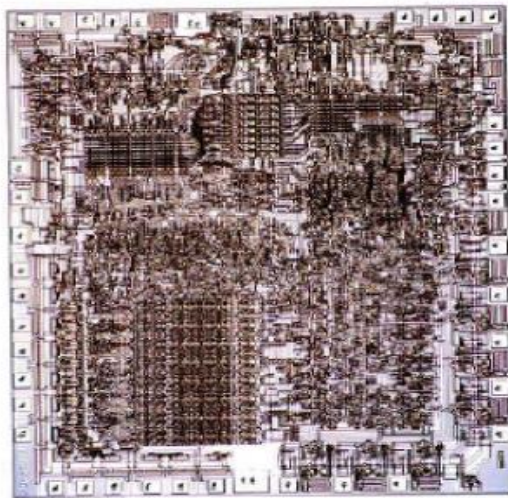


Intel founders (circa 1978)

Image: Intel

IBM INTRODUCED A PERSONAL COMPUTER IN 1981

- First IBM computer used the Intel 8088 processor
- 8088 contained 29,000 transistors •
- Used Operating System (MS-DOS) designed by Microsoft



Intel 8088

Image: Intel



IBM PC

Image: Intel

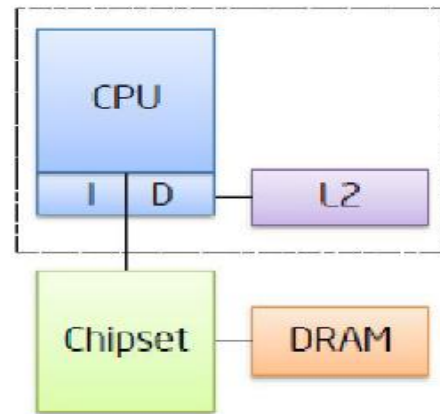
MICROPROCESSOR EVOLUTION

- Human hair- 100 μm across
- 4004 transistors- 10 μm across
- Pentium 4 transistors- 0.13 μm across
- Core2Duo- 65 nm (0.065 μm) across
- Latest i7- 10nm across, AMD Ryzen4 5nm
- Future cores (2024)- 3nm across

COMPUTER COMPONENTS

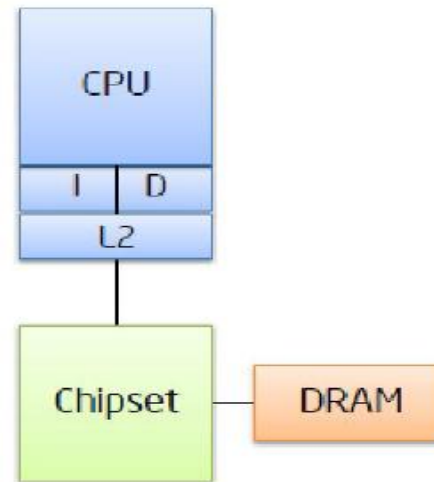
- Microprocessor/Microcontroller — performs all computations
- Cache (On Chip) — fast memory which holds current data and program
- Main memory (Off Chip) — larger DRAM memory contains more data
- Chipset — controls communication between components
- Motherboard/Development Board — circuit board which holds all the above components
- Peripherals — controls added computer accessories
- Displays- 7Segments, LCD, LED

CPU AND MEMORY EVOLUTION



Pentium II

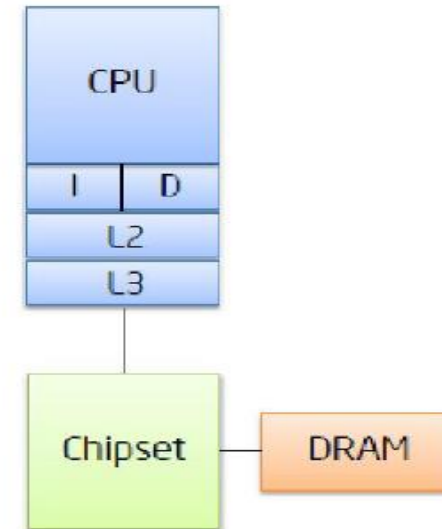
Separate bus to L2
cache in same
package



Pentium III

L2 cache on-die

More no. of cores

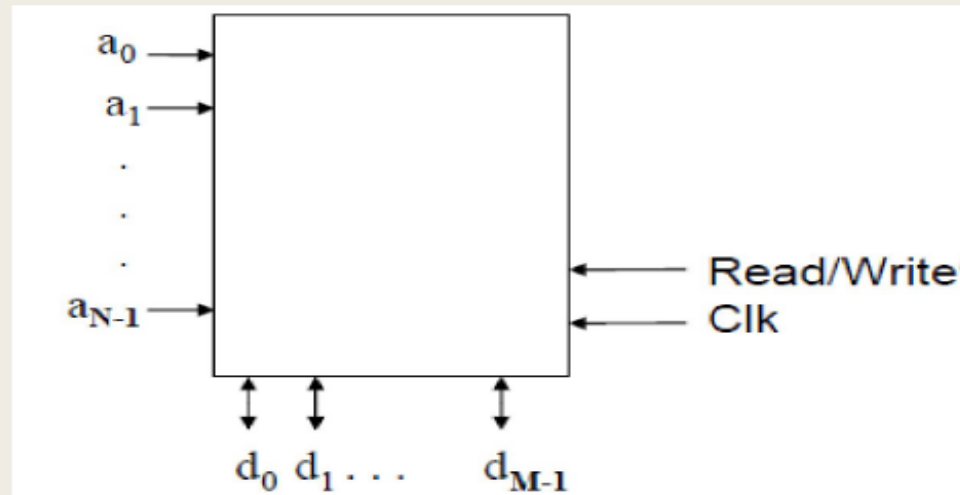


Core i7

L3 cache on-die

Memory

- Another common subsystem in microprocessors is MEMORY, with multiple address lines and multiple data lines:



- The above has N address lines and M data lines. The capacity of this memory system is ???

Memory Types: ROM and RAM

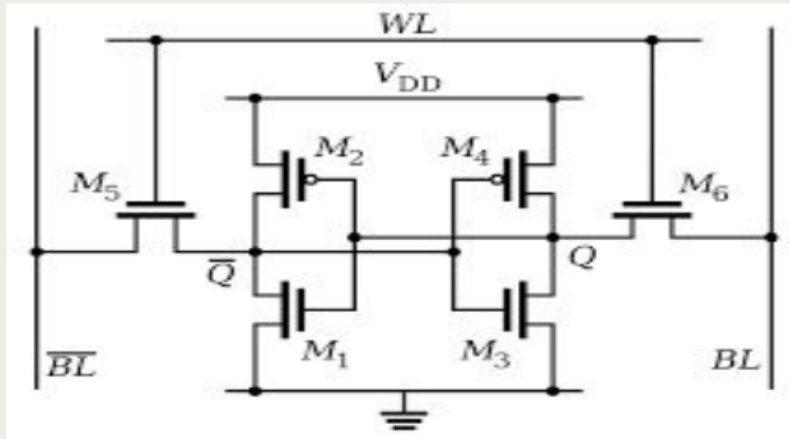
■ Random Access Memory

- **Dynamic Random Access Memory (DRAM):** *periodic refresh is required to maintain the contents of a DRAM chip*
- **Static Random Access Memory (SRAM):** *no periodic refresh is required. Always more predictable and usually faster than DRAM.*

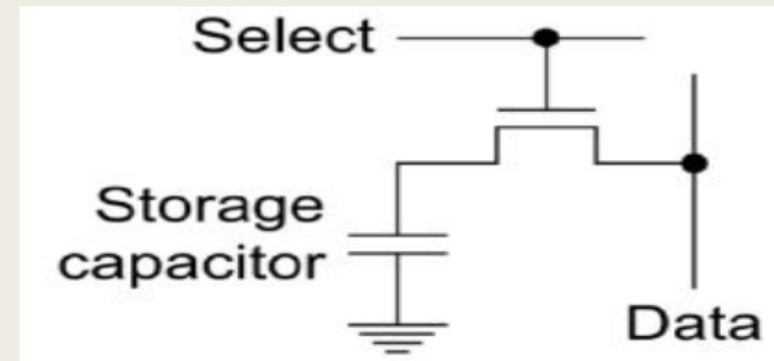
■ Read-Only Memory

- **Mask-programmed read-only memory (MROM):** *programmed when being manufactured*
- **Programmable read-only memory (PROM):** *the memory chip can be programmed by the end user*

Memory: SRAM and DRAM



SRAM single cell



DRAM single cell

EPROM, EEPROM, FLASH

- **Erasable Programmable ROM (EPROM)**
 1. Electrically programmable many times
 2. Erased by ultraviolet light (through a window)
 3. Erasable in bulk (whole chip in one erasure operation)
- **Electrically Erasable Programmable ROM (EEPROM)**
 1. Electrically programmable many times
 2. Electrically erasable many times
 3. Can be erased one location, one row, or whole chip in one operation
- **Flash Memory**
 1. Electrically programmable many times
 2. Electrically erasable many times
 3. Can only be erased in bulk

Memory Addressing

■ Memory Addressing

- Memory consists of a sequence of directly addressable locations.
- A location is referred to as an information unit.
- A memory location can be used to store data and instructions.
- A memory location has two components: an address and its contents.

Address



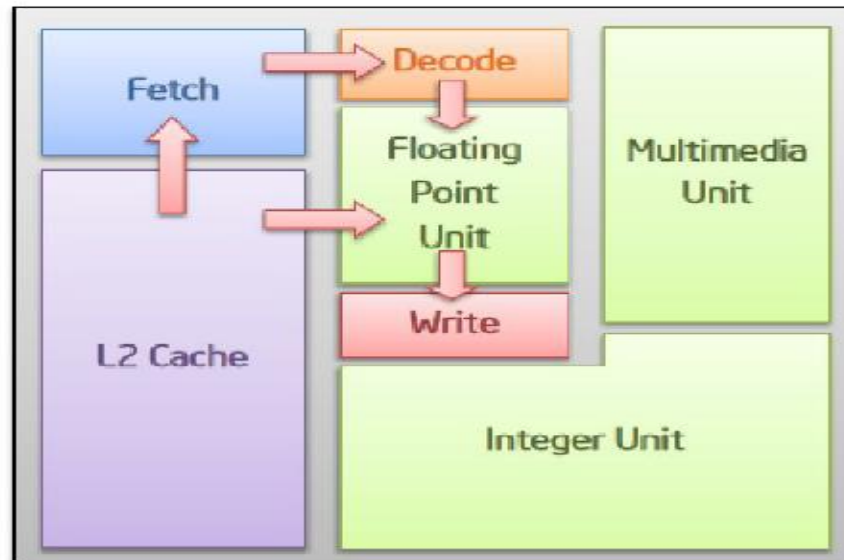
Contents

The components of a memory location

Architecture Basics

- **Computer Programs** - Computer programs (programs) are known as software - A program is a sequence of machine instructions
- **Machine Instruction** - A sequence of binary digits that can be executed by the processor - Hard to understand, program, and debug for human being - E.g. in the ARM Cortex M4,
 - instruction **0010** 0010 0110 **0100**
 - *How many bytes long is this instruction? Instruction is also 0x2264.*
- **Assembly Language** - Defined by machine instructions - An assembly instruction is a mnemonic representation of a machine instruction,
 - *ADD R2 R1 R0 represents 0001100 010 001 000 = 0x1888 (hex format)*

CPU stages



Fetch Unit gets the next instruction from the cache.

Decode Unit determines type of instruction.

Instruction and data sent to **Execution Unit**.

Write Unit stores result.



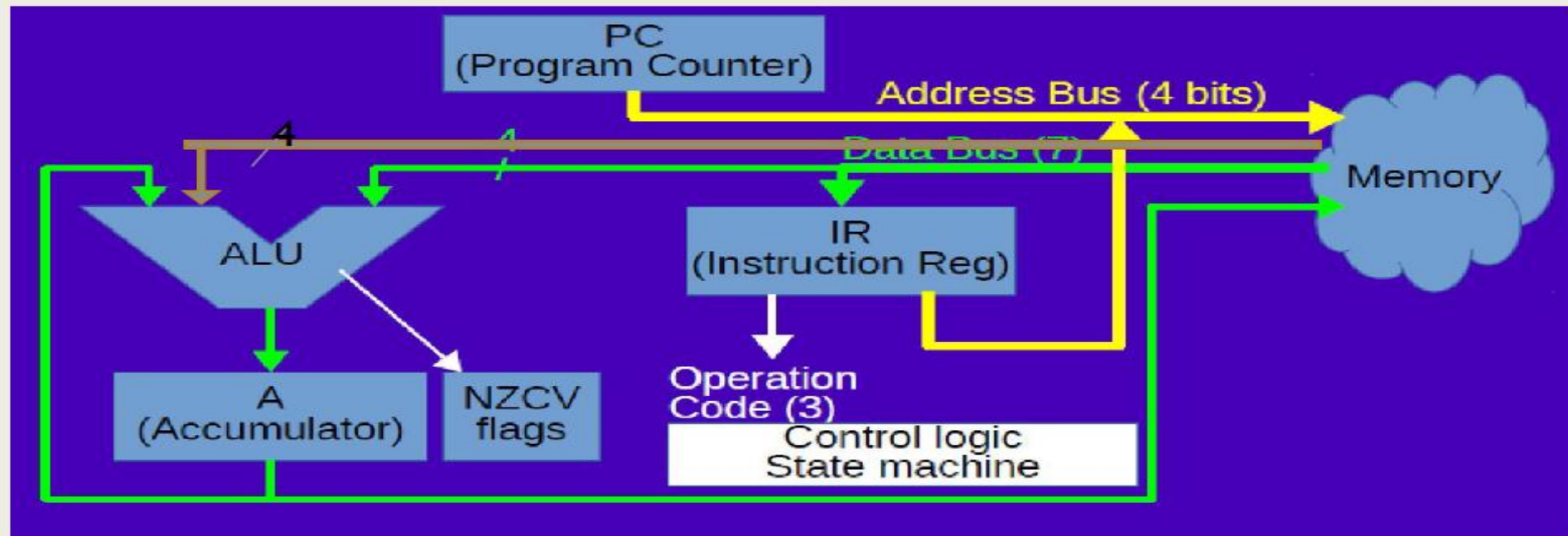
Architecture and Operations

- **Program counter:** register PC “points to” (i.e. contains the address of) the memory location containing the next instruction to be executed.
- **Fetch:** Using the current value of PC, read the next instruction to execute from memory. Store this data in the instruction register, IR.
- Note that IR is an internal register, not accessible with computer instructions.

CPU stages (Pipelines)

- **Decode :** Determine which instruction is to be executed using the contents of the IR.
- **Execute :** Activate appropriate control lines in processor sequentially to perform the action specified by the Instruction at particular clock ticks.
- **Write:** Write the output to the Data Memory

A SIMPLE CPU: PC, ALU, ACCUMULATOR, IR, MEMORY, BUSES, CONTROL LOGIC



INSTRUCTIONS AND PROGRAMMING

- Instructions are the most basic actions the processor can process:
 - `ADD AX, BX` # Add value AX to BX and store in AX
 - `CMP AX, 5` # Compare value in AX to 5
 - `JE 16` # Jump ahead 16 bytes if comparison was equal
- Instruction Set Architecture (ISA) defines all the instructions of an architecture (e.g. ARM, X86)
- High level programming languages (C, C++, Java) allow many processor instructions to be written simply:
 - `if (A + B = 5)`
 - `then... #Jump if sum of A and B is 5`
- Every program (e.g. C) must be converted to the processor instructions (e.g. binaries) of the computer it will be run on

A Instruction Stages Example

- Let's execute the instruction ADD R4,R3,R2
- It is a 32 bit instruction located in Instruction Memory (I-cache)
- This instruction means: add contents of register R2 to the contents of register R3, and store the results into register R4.
- Here's what happens: **Fetch** instruction, **decode**, and read from register addresses 0x00011 (R3) and 0x00010 (R2)
- Connect register R2 to input 1 of the ALU and Connect register R3 to input 2 of the ALU.
- Set controls to ALU to do a 32-bit add operation (**Execute**)
- Store the output of the ALU into register R4 (i.e. connect R4 to output of ALU) (**Write**)
- Increment the program counter and store that value back in the program counter: $[PC] = [PC] + 4$

MICROPROCESSOR ARCHITECTURE

- Microprocessor (Computer) architecture is defined by the instructions (ISA) a processor can execute
- • Programs (e.g. C) written for one processor can run on any other processor of the same architecture
- Current architectures include:
 - IA32 also known as x86
 - IA64 also known as x86_54 (Intel,AMD)
 - ARM (Qualcomm,Apple)
 - PowerPC (IBM)
 - MIPS
 - RISC V (Open Source)
 - Xtensa (ESP32)

Simulator

LEG8 Simulator: ARM University Education Program

<https://github.com/arm-university/Graphical-Micro-Architecture-Simulator>

Browser Based

Needs to be downloaded

CPULator: University of Toronto

<https://cpulator.01xz.net/doc/>

Browser Based

No need to be downloaded